Huawei Access Node HSR-ACC-74 Optical Transport Failure SOP (ALARM_1962)

Node: HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_1962

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a minor optical transport failure on Huawei Access Node HSR-ACC-74, located at HSR Layout Access. The failure domain is categorized under Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss). The Huawei Access Node is an NE20E-X8, with a role of Access and a CLI dialect of Huawei. The network architecture is based on the Huawei's Optical Transport Network (OTN) protocol, which utilizes the OTU4 (Optical Transport Unit level 4) framing structure over DWDM (Dense Wavelength Division Multiplexing) links. The OTU4 frames are transmitted over the fiber infrastructure, which can be susceptible to degradation or cuts, resulting in Rx/Tx power loss and ultimately, the ALARM_1962 alarm code. To mitigate this issue, it is crucial to first identify the root cause of the failure, which could be related to the SFP/XFP degradation, fiber cuts, or other environmental factors affecting the fiber infrastructure. Once the root cause is identified, the corresponding resolution steps can be taken to restore the normal functioning of the Huawei Access Node.

Pre-requisite Checks

- Ensure remote hands are dispatched to HSR Layout Access for physical access to the Huawei Access Node HSR-ACC-74.

- Verify the status of the redundant link (if present) to ensure it is functioning correctly.

- Check the environmental conditions (temperature, humidity, etc.) surrounding the Huawei Access Node to rule out any external factors affecting the fiber infrastructure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Environmental factors affecting the fiber infrastructure (temperature, humidity, etc.)	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface, Expected Output: Optical Transport Interface Information
---------------------------------------- Interface Name : 1/1/1 Interface Type : OTU4 Interface Status : Down , Description: This command displays the optical transport interface information, including the interface name, type, and status. The expected output indicates that the interface is down, which could be related to the
```

```
failure.

Command: display optical-transport interface optical-parameters,
Expected Output: Optical Transport Interface Optical Parameters
---------------------------------------------------------- Interface Name
: 1/1/1 Rx Power : -30.5 dBm Tx Power : -10.5 dBm , Description: This
command displays the optical transport interface optical parameters,
including the receive and transmit power. The expected output
indicates that the receive power is lower than the expected value,
which could be related to the failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transport interface 1/1/1, Description: Replace the SFP/XFP module on the Huawei Access Node HSR-ACC-74 with a new one to resolve the SFP/XFP degradation issue.

Step 2: Step: Verify the fiber infrastructure, Command: display optical-transport interface optical-parameters, Description: Verify the fiber infrastructure by checking the receive and transmit power values. If the values are within the expected range, it could indicate that the fiber cuts issue is resolved.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output: Optical
Transport Interface Information
---------------------------------------- Interface Name : 1/1/1
Interface Type : OTU4 Interface Status : Up , Description: This
command verifies that the optical transport interface is up and
functioning correctly.

Command: display optical-transport interface optical-parameters,
Expected Output: Optical Transport Interface Optical Parameters
---------------------------------------------------------- Interface Name
: 1/1/1 Rx Power : -20.5 dBm Tx Power : -10.5 dBm , Description: This
command verifies that the receive and transmit power values are within
the expected range.
```

Roll-back Procedures

- Step: Undo the replacement of the SFP/XFP module, Command: undo replace optical-transport interface 1/1/1, Description: Undo the replacement of the SFP/XFP module by executing the undo command to restore the previous configuration.